

Bansilal Ramnath Agarwal Charitable Trust's
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(An autonomous Institute of Savitribai Phule Pune University)



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Engineering

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Title:- *handwritten title of project 7 2021*
Create a database with appropriate constraints using DDL & populate / modify it with the help of DML

Description: In this assignment, we applied various SQL operations on two tables — Available_Equipments and Exercises — demonstrating both DML and DDL commands. For Available_Equipments, we created the table with a primary key, inserted ten records, updated and deleted specific entries, added and modified a column, dropped it, and renamed both a column and the table. For Exercises, which has a foreign key referencing Available_Equipments, we inserted ten linked exercise records, performed updates and deletions, added and modified a column, removed it, and renamed a column and the table. These steps covered key SQL functions for managing and modifying relational databases.

Screenshots/Output:

Available Equipments Table:

The screenshot displays a database management interface. At the top, a tab labeled 'Equipment Inventory Setup' is active. Below it, a SQL editor shows the following code:

```
1 CREATE TABLE Available_Equipments (  
2   equipment_id SERIAL PRIMARY KEY,  
3   equipment_name TEXT NOT NULL  
4 );  
5
```

Below the SQL editor, the table 'public.available equipments' is selected. The table structure is shown with columns 'equipment_id' (int4) and 'equipment_name' (text). A '+ Table Creation' button is visible. The table is currently empty, with a message 'This table is empty' and an 'Import data from CSV' button. The bottom of the interface shows pagination controls: 'Page 1 of 1', '100 rows', and '0 records'. There are also 'Refresh', 'Data', and 'Definition' buttons.

Equipment Inventory Addition

```
1 INSERT INTO Available_Equipments (equipment_name) VALUES
2 ('Barbell'),
3 ('Squat Rack'),
4 ('Cable Machine'),
5 ('Dumbbells'),
6 ('Resistance Bands'),
7 ('Kettlebell'),
8 ('Treadmill'),
9 ('Stationary Bike'),
10 ('Pull-up Bar'),
11 ('Medicine Ball');
12
```

Insert 10 Rows

public.available equipments

Filter

Sort

Insert

RLS disabled

Role postgres

Realtime off

API Docs

	equipment_id	equipment_name	
	1	Barbell	
	2	Squat Rack	
	3	Cable Machine	
	4	Resistance Bands	
	5	Kettlebell	
	6	Treadmill	
	7	Stationary Bike	
	8	Pull-up Bar	
	9	Medicine Ball	
	10	Dumbbells	

Page 1 of 1

100 rows

10 records

Refresh

Data

Definition

Untitled query

```
1 UPDATE Available_Equipments
2 SET equipment_name = 'Adjustable Dumbbells'
3 WHERE equipment_id = 4;
4
```

Update

public.available equipments

Filter

Sort

Insert

RLS disabled

Role postgres

Realtime off

API Docs

☐	equipment_id	equipment_name	
☐	1	Barbell	
☐	2	Squat Rack	
☐	3	Cable Machine	
☐	4	Adjustable Dumbbells	
☐	5	Kettlebell	
☐	6	Treadmill	
☐	7	Stationary Bike	
☐	8	Pull-up Bar	
☐	9	Medicine Ball	
☐	10	Dumbbells	

Page 1 of 1

100 rows

10 records

Refresh

Data

Definition

Update Equipment Name

Equipment Removal

1 DELETE FROM Available_Equipments

2 WHERE equipment_id = 10;

3

Delete

☐	equipment_id	equipment_name	
☐	1	Barbell	
☐	2	Squat Rack	
☐	3	Cable Machine	
☐	4	Adjustable Dumbbells	
☐	5	Kettlebell	
☐	6	Treadmill	
☐	7	Stationary Bike	
☐	8	Pull-up Bar	
☐	9	Medicine Ball	

Page 1 of 1

100 rows

9 records

Refresh

Data

Definition

Add Equipment Category Column

1 ALTER TABLE Available_Equipments

2 ADD COLUMN category TEXT;

3

Add a column

public.available equipments

Filter

Sort

Insert

RLS disabled

Role postgres

Realtime off

API Docs

	equipment_id	equipment_name	category	
	1	Barbell	NULL	
	2	Squat Rack	NULL	
	3	Cable Machine	NULL	
	4	Adjustable Dumbbells	NULL	
	5	Kettlebell	NULL	
	6	Treadmill	NULL	
	7	Stationary Bike	NULL	
	8	Pull-up Bar	NULL	
	9	Medicine Ball	NULL	

Page 1 of 1

100 rows

9 records

Refresh

Data

Definition

Modify Equipment Category Field Length

1 ALTER TABLE Available_Equipments

2 ALTER COLUMN category TYPE VARCHAR(50);

3

Modify column type

public.available equipments

Filter Sort Insert RLS disabled Role postgres Realtime off API Docs

	equipment_id int4	equipment_name text	category varchar	+
☐	1	Barbell	NULL	
☐	2	Squat Rack	NULL	
☐	3	Cable Machine	NULL	
☐	4	Adjustable Dumbbells	NULL	
☐	5	Kettlebell	NULL	
☐	6	Treadmill	NULL	
☐	7	Stationary Bike	NULL	
☐	8	Pull-up Bar	NULL	
☐	9	Medicine Ball	NULL	

Page 1 of 1 100 rows 9 records Refresh Data Definition

Modify Equipment Category Field Length

Untitled query

```

1 ALTER TABLE Available_Equipments
2 DROP COLUMN category;
3

```

Drop a column

public.available equipments

Filter Sort Insert RLS disabled Role postgres Realtime off API Docs

	equipment_id int4	equipment_name text	+
☐	1	Barbell	
☐	2	Squat Rack	
☐	3	Cable Machine	
☐	4	Adjustable Dumbbells	
☐	5	Kettlebell	
☐	6	Treadmill	
☐	7	Stationary Bike	
☐	8	Pull-up Bar	
☐	9	Medicine Ball	

Page 1 of 1 100 rows 9 records Refresh Data Definition

Rename Equipment Column

```
1 ALTER TABLE Available_Equipments
2 RENAME COLUMN equipment_name TO name;
3
```

Rename Column

public.available equipments

Filter Sort Insert

RLS disabled Role postgres Realtime off API Docs

	equipment_id	name	
☐	1	Barbell	
☐	2	Squat Rack	
☐	3	Cable Machine	
☐	4	Adjustable Dumbbells	
☐	5	Kettlebell	
☐	6	Treadmill	
☐	7	Stationary Bike	
☐	8	Pull-up Bar	
☐	9	Medicine Ball	

Page 1 of 1 100 rows 9 records Refresh Data Definition

Rename Equipment Table

```
1 ALTER TABLE Available_Equipments
2 RENAME TO Gym_Equipments;
3
```

Rename Table

public.gym equipments

Filter

Sort

Insert

RLS disabled

Role postgres

Realtime off

API Docs

☐	equipm...	name	text	
☐	1	Barbell		
☐	2	Squat Rack		
☐	3	Cable Machine		
☐	4	Adjustable Dumbbells		
☐	5	Kettlebell		
☐	6	Treadmill		
☐	7	Stationary Bike		
☐	8	Pull-up Bar		
☐	9	Medicine Ball		

Page 1 of 1

100 rows

9 records

Refresh

Data

Definition

Exercises Table:

Exercise Tracking Table

```
1 CREATE TABLE Exercises (  
2   exercise_id SERIAL PRIMARY KEY,  
3   exercise_name TEXT NOT NULL,  
4   muscle_group TEXT,  
5   equipment_id INT NOT NULL,  
6   FOREIGN KEY (equipment_id) REFERENCES Available_Equipments(equipment_id)  
7   ON DELETE CASCADE  
8 );  
9
```

Table Creation

exercises

Filter

Sort

Insert

RLS disabled

Role postgres

Realtime off

API Docs

exercise_id

int4

exercise_name

text

muscle_group

text

equipment_id

int4

This table is empty

Import data from CSV

or drag and drop a CSV file here

Page 1 of 1

100 rows

0 records

Refresh

Data

Definition

Exercise Data Population

```
1 INSERT INTO Exercises (exercise_name, muscle_group, equipment_id) VALUES  
2 ('Bench Press', 'Chest', 1),  
3 ('Squat', 'Legs', 2),  
4 ('Deadlift', 'Back', 1),  
5 ('Bicep Curl', 'Biceps', 4),  
6 ('Tricep Pushdown', 'Triceps', 3),  
7 ('Shoulder Press', 'Shoulders', 4),  
8 ('Lat Pulldown', 'Back', 3),  
9 ('Leg Press', 'Legs', 2),  
10 ('Chest Fly', 'Chest', 4),  
11 ('Plank with Bands', 'Core', 5);  
12
```

Insert 10 Rows

public.exercises

FilterSortInsertRLS disabledRole postgresRealtime offAPI Docs

☐	exercise_id int4	exercise_name text	muscle_group text	equipment_id int4	
☐	1	Bench Press	Chest	1	→
☐	2	Squat	Legs	2	→
☐	3	Deadlift	Back	1	→
☐	4	Bicep Curl	Biceps	4	→
☐	5	Tricep Pushdown	Triceps	3	→
☐	6	Shoulder Press	Shoulders	4	→
☐	7	Lat Pulldown	Back	3	→
☐	8	Leg Press	Legs	2	→
☐	9	Chest Fly	Chest	4	→
☐	10	Plank with Bands	Core	5	→

Page 1 of 1100 rows10 recordsRefreshDataDefinition

Update Exercise Muscle Group

1 UPDATE Exercises
2 SET muscle_group = 'Upper Chest'
3 WHERE exercise_id = 1;
4

Update

public.exercises

FilterSortInsertRLS disabledRole postgresRealtime offAPI Docs

☐	exercise_id int4	exercise_name text	muscle_group text	equipment_id int4	
☐	1	Bench Press	Upper Chest	1	→
☐	2	Squat	Legs	2	→
☐	3	Deadlift	Back	1	→
☐	4	Bicep Curl	Biceps	4	→
☐	5	Tricep Pushdown	Triceps	3	→
☐	6	Shoulder Press	Shoulders	4	→
☐	7	Lat Pulldown	Back	3	→
☒	8	Leg Press	Legs	2	→
☐	9	Chest Fly	Chest	4	→
☐	10	Plank with Bands	Core	5	→

Page 1 of 1100 rows10 recordsRefreshDataDefinition

Remove Exercise #10

1

DELETE FROM Exercises

2

WHERE exercise_id = 10;

3

Delete

public.exercises

Filter

Sort

Insert

RLS disabled

Role postgres

Realtime off

API Docs

	exercise_id	exercise_name	muscle_group	equipment_id	
	1	Bench Press	Upper Chest	1	
	2	Squat	Legs	2	
	3	Deadlift	Back	1	
	4	Bicep Curl	Biceps	4	
	5	Tricep Pushdown	Triceps	3	
	6	Shoulder Press	Shoulders	4	
	7	Lat Pulldown	Back	3	
	8	Leg Press	Legs	2	
	9	Chest Fly	Chest	4	

Page 1 of 1

100 rows

9 records

Refresh

Data

Definition

Add Exercise Difficulty Level

1

ALTER TABLE Exercises

2

ADD COLUMN difficulty_level TEXT;

3

Add a column

public.exercises						
Filter	Sort	Insert		RLS disabled	Role postgres	Realtime off API Docs
	exercise_id	exercise_name	muscle_group	equipment_id	difficulty_level	
	1	Bench Press	Upper Chest	1	NULL	
	2	Squat	Legs	2	NULL	
	3	Deadlift	Back	1	NULL	
	4	Bicep Curl	Biceps	4	NULL	
	5	Tricep Pushdown	Triceps	3	NULL	
	6	Shoulder Press	Shoulders	4	NULL	
	7	Lat Pulldown	Back	3	NULL	
	8	Leg Press	Legs	2	NULL	
	9	Chest Fly	Chest	4	NULL	

Modify Exercise Difficulty Format

+

1 ALTER TABLE Exercises

2 ALTER COLUMN difficulty_level TYPE VARCHAR(50);

3

Modify column type

public.available equipments

public.exercises

+

Filter

Sort

Insert

RLS disabled

Role postgres

Realtime off

API Docs

	exercise_id int4	exercise_name text	muscle_group text	equipment_id int4	difficulty_level varchar	
	1	Bench Press	Upper Chest	1	NULL	
	2	Squat	Legs	2	NULL	
	3	Deadlift	Back	1	NULL	
	4	Bicep Curl	Biceps	4	NULL	
	5	Tricep Pushdown	Triceps	3	NULL	
	6	Shoulder Press	Shoulders	4	NULL	
	7	Lat Pulldown	Back	3	NULL	
	8	Leg Press	Legs	2	NULL	
	9	Chest Fly	Chest	4	NULL	

Page 1 of 1

100 rows

9 records

Refresh

Data

Definition

Remove Difficulty Level

1

ALTER TABLE Exercises

2

DROP COLUMN difficulty_level;

3

Drop a column

public.exercises

FilterSortInsert

RLS disabledRole postgresRealtime offAPI Docs

	exercise_id int4	exercise_name text	muscle_group text	equipment_id int4	
	1	Bench Press	Upper Chest	1	→
	2	Squat	Legs	2	→
	3	Deadlift	Back	1	→
	4	Bicep Curl	Biceps	4	→
	5	Tricep Pushdown	Triceps	3	→
	6	Shoulder Press	Shoulders	4	→
	7	Lat Pulldown	Back	3	→
	8	Leg Press	Legs	2	→
	9	Chest Fly	Chest	4	→

Page 1 of 1100 rows9 recordsRefreshDataDefinition

Rename Exercise Column

1

ALTER TABLE Exercises

2

RENAME COLUMN exercise_name TO name;

3

Rename Column

public.exercises

FilterSortInsertRLS disabledRole postgresRealtime offAPI Docs

	exercise_id int4	muscle_group text	equipment_id int4	name text	
	1	Upper Chest	1	Bench Press	
	2	Legs	2	Squat	
	3	Back	1	Deadlift	
	4	Biceps	4	Bicep Curl	
	5	Triceps	3	Tricep Pushdown	
	6	Shoulders	4	Shoulder Press	
	7	Back	3	Lat Pulldown	
	8	Legs	2	Leg Press	
	9	Chest	4	Chest Fly	

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Rename Exercises Table

ALTER TABLE Exercises
RENAME TO Gym_Exercises;

Rename Table

public.gym_exercises

FilterSortInsertRLS disabledRole postgresRealtime offAPI Docs

	exerci... i...	name text	muscle_group text	equipm... i...	
	1	Bench Press	Upper Chest	1	
	2	Squat	Legs	2	
	3	Deadlift	Back	1	
	4	Bicep Curl	Biceps	4	
	5	Tricep Pushdown	Triceps	3	
	6	Shoulder Press	Shoulders	4	
	7	Lat Pulldown	Back	3	
	8	Leg Press	Legs	2	
	9	Chest Fly	Chest	4	

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